

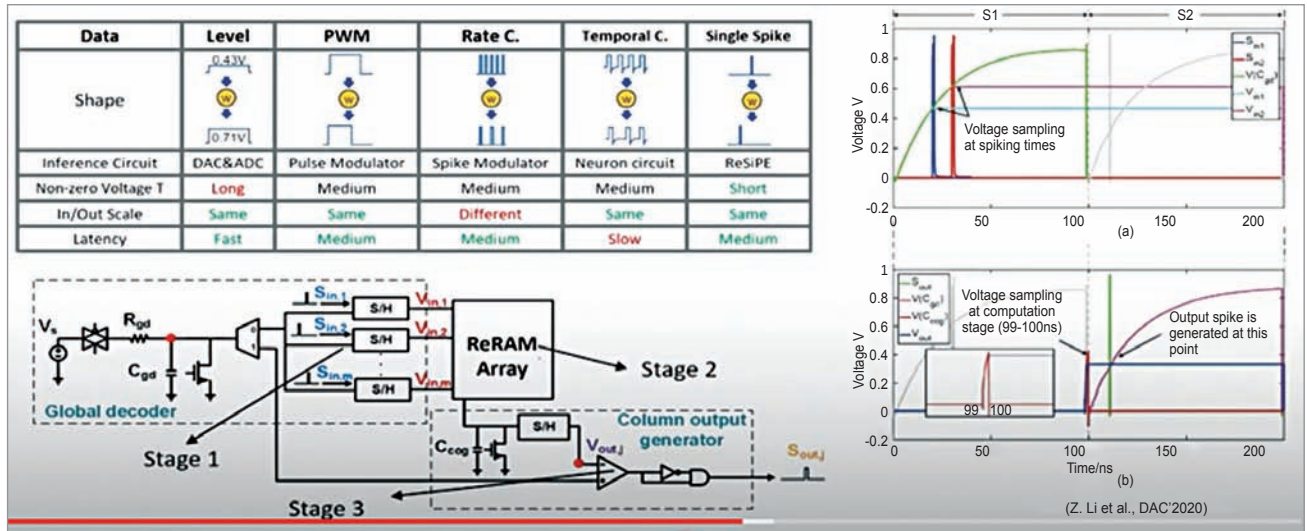


Figure 11: ReRAM based processing engine: ReSiPE

We consider the scaling factor ‘s’ in Ws as an independent trainable variable. This way we are able to enable gradient based training of bit representation. During the training we allow each bit to take a continuous value between zero and two.

Memristor—rebirth of neuromorphic circuits: ReRam, memristor, or metal-oxide resistive random access memory is essentially for simple two-terminal devices. When we apply voltage or pass current through the devices, their resistance level could change. It is often also called the Programmable Resistance with analogue states.

The spike based neuromorphic design: We have moved to spike based neuromorphic design and minimisation of power consumption. It is very similar to the previous structure, but here we use the frequency of the spike to represent the input data amplitude. So, the information is encoded in the spikes instead of analogue voltage. Accordingly, fire circuits are used to produce output spikes along the bit line levels.

We measure the output as impulse number versus the input as the summation of the weighted inputs. The initial expectation was that these will follow a straight

line like proportional relations, but the measurements show that it could get saturated. This is mainly because of the operation of the IFC (integrate-and-fire case), which have an upper bound.

Processing engine with in-situ neuron activation (ISNA): We integrate 64-kilobit RRAM arrays in this engine. The bottom part of this device is so much like an irregular memory array. On the top part we add our neural processing components (see Figure 8).

Spike conversion: The design still uses capacitors to generate spikes. Over here (see Figure 9), we use the output of the op-amp to form a positive feedback to accelerate the responses from the current amplifier. We use the current mirror to scale down the bit-line current, because in the real

network applications, the current can either be several microamperes or several milliamperes—especially at the early stage when the device is not very stable and could result in a larger range of resistance.

In-situ nonlinear activation function: In the following design (see Figure 10) we have two knobs. The reference voltage can control the read voltage of the bit lines while the threshold voltage is used to control the capacitors along with charging and discharging speed. With these distant knobs we are able to slightly tune the shape of neural activation functionalities.

Configurable activation precision: We measured our chips on the application level and explored trade-offs between high precision weight and activation versus improved performance. Such exploration is useful for system-level, fine-grained optimisation and here are the results.

As mentioned earlier, the designs were fabricated by 150-nanometer complementary metal-oxide semiconductor (CMOS) and HfO RRAM devices. They contain 64-kilobit RRAM devices. The energy efficiency is also high at 0.257 pico joules (pJ/MAC).

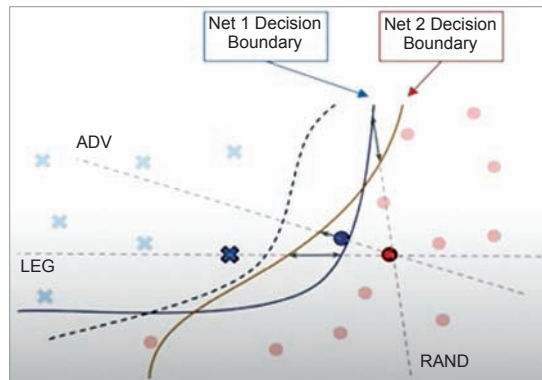


Figure 12: Transfer-based attacks